



1. Heat flows from **hot** to **cold**.
2. Heat flows faster when the temperature difference is larger.
3. More heat enters when the surface area is larger.
4. Temperature rises in proportion to the heat added.
5. Temperature rise is smaller when mass is larger.

1, 2, 3 $\rightarrow$

$$\Delta H = k \times A \times \frac{\Delta T}{D} \times \Delta t$$

(heat added) $\downarrow$ (surface area) (temperature "slope") (spreading time)

(thermal conductivity)

4, 5 $\rightarrow$

$$\Delta H = c \times M \times \Delta T$$

(specific heat)

Put it together:

$$k A \frac{\cancel{\Delta t}}{D} \Delta t = c M \cancel{\Delta t}$$

$$\Delta t = \frac{c M D}{k A}$$

$$* \text{ Density } = \rho = \frac{\text{mass}}{\text{volume}} \sim \frac{M}{D^3}$$

$$* \text{ Area } = A \sim D^2$$

$$\Delta t \sim \frac{c M D}{k D^2} \times \frac{D}{D} \sim \frac{c \rho}{k} D^2$$

$$D \sim \left(\frac{M}{\rho}\right)^{1/3}$$

$$\Delta t \sim \frac{c \rho^{1/3}}{k} \underline{\underline{M^{2/3}}}$$

$$\frac{\Delta t(1016)}{\Delta t(116)} = 4.6$$